

新疆省政府主席
此項官票每紅錢肆百文
湘平銀壹兩民間完糧
解繳公款一律通用
造定即治罪此示
民國十九年

如有偽稅作示
日

نېنگىيەنىڭ سىياسىي ھۆكۈمىتى
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قۇتۇپتۇر ساراڭنىڭ قىزىنىڭ قەۋمى
نىڭ ئىكەنلىكىنى باھانىسىدا قىلغان
بالغاننىڭ جەھەتتىن ئۆزىنىڭ
دېيىپ چىققانلىقىنى
ھۆكۈم ئۆزگەرتىشنى بىلەن

asking those who feel it

local and Indigenous knowledge (ILK)
on climate change

Day 3 (7 December)

Jakob Steiner

Course content

5 December (1400 – 1800)

Pre course evaluation - general introduction and history – guest lecture (Arctic)- introduction to writing assignment

6 December (0900 – 1700)

Guest lecture (Himalaya) – Climate Science– interactive team work - International efforts to integrate ILK

7 December (0900 – 1700)

ILK governance and ethics – integrating ILK with conventional data - interactive team work - group presentation and discussion

7 December / Day 3

Welcome to our third session 😊

Learning Objectives: *Understand and be able to explain how Indigenous and local knowledge is today included in global assessments of our state of knowledge of the environment and climate and where gaps are. Understand how such assessments are compiled and what larger aim the inclusion of Indigenous and local knowledge could achieve, if it is included.*

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7 December / Day 3

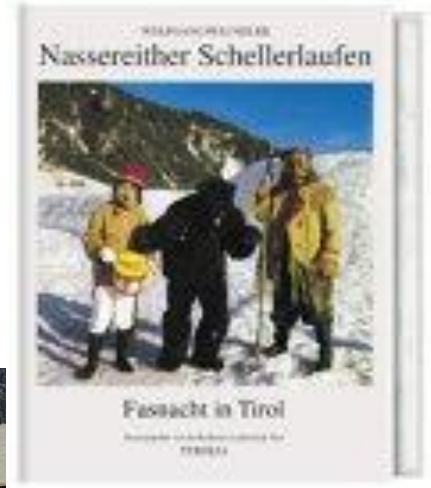
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Governance, Ownership and Ethics



Governance, Ownership and Ethics

Acknowledgment that environmental problems are to a large degree related to failures in 'management' or 'governing' our resources.

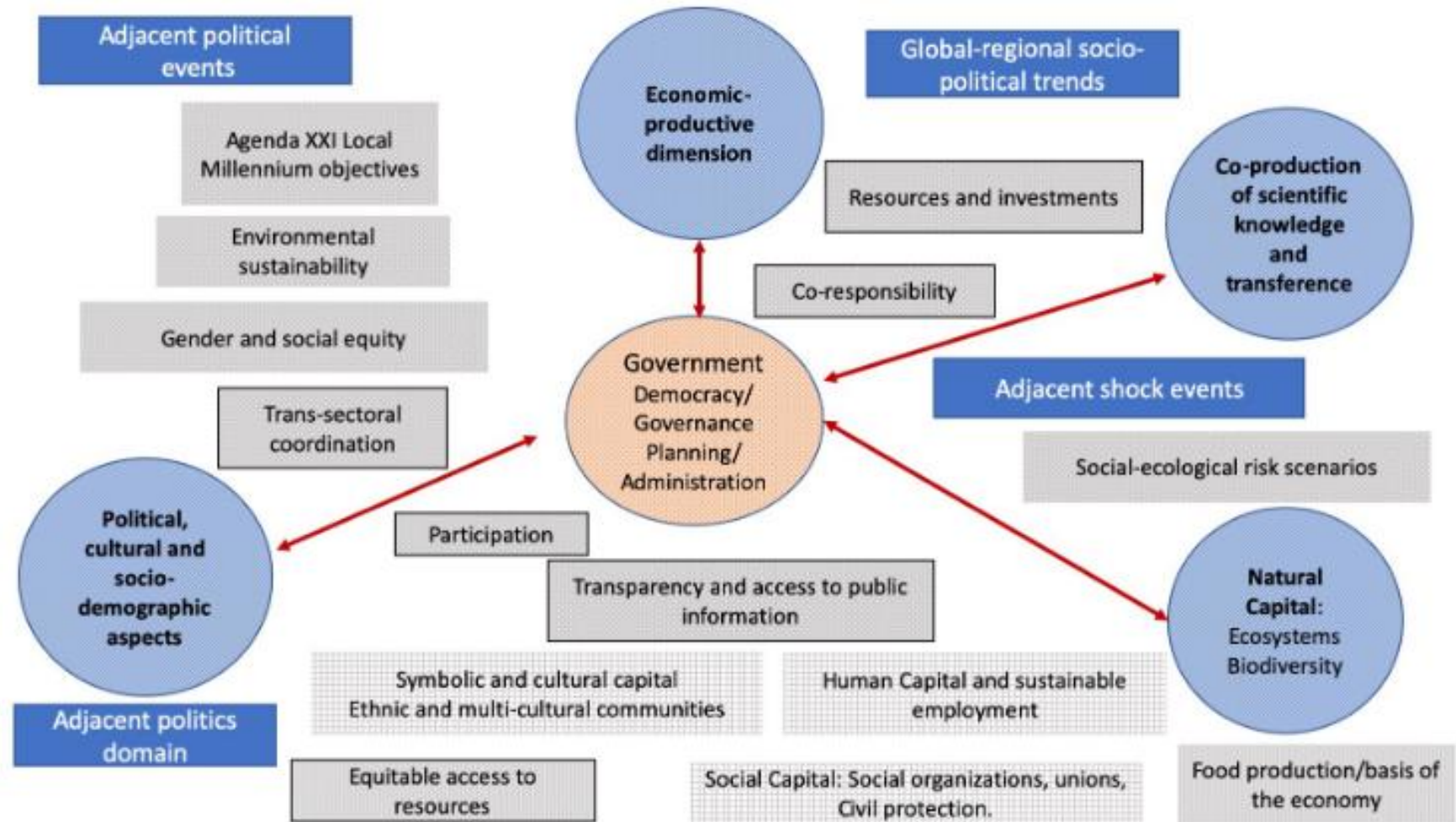
Especially developed related to water (water governance)

Inclusion of Indigenous and local knowledges exists in Literature (e.g. Holley et al., 2012, '*New Environmental Governance*') and as a UN Declaration (UNDRIP, 2007)

Co-production of knowledge as part of governance

adaptive-
collaborative
knowledge production

No focus on
Indigenous Peoples or
ownership specifically



Holley et al. (2012, 2019) **Figure 1.** Pillars, fundamental principles and capitals for the development of an environmental governance (Modified from Arnouts *et al.* 2012).

United National Declaration on the Rights of Indigenous Peoples

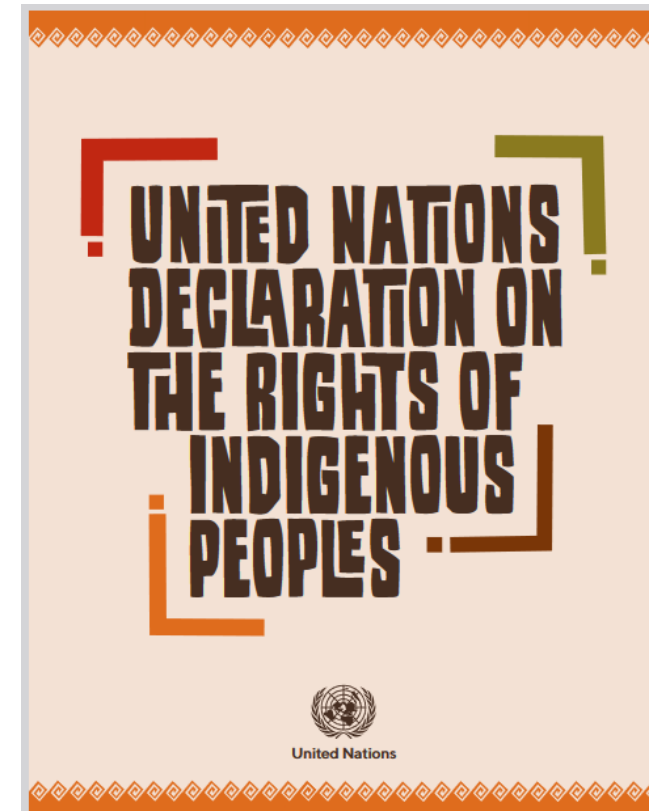
No generally accepted definition and no definition under United Nations or International Law

- *self-identification*
- *cultural difference* from other groups in a state
- a *special relationship* with their *traditional territory*
- experience of *subjugation* and discrimination under a *dominant cultural model*

'Knowledge' only mentioned 4 times

Article 31

1. Indigenous peoples have the right to maintain, control, protect and develop their cultural heritage, traditional knowledge and traditional cultural expressions, as well as the manifestations of their sciences, technologies and cultures, including human and genetic resources, seeds, medicines, knowledge of the properties of fauna and flora, oral traditions, literatures, designs, sports and traditional games and visual and performing arts. They also have the



Nagoya Protocol 2010

Recognizing that an innovative solution is required to address the fair and equitable sharing of benefits derived from the utilization of genetic resources and traditional knowledge associated with genetic resources that occur in transboundary situations or for which it is not possible to grant or obtain prior informed consent,

Article

7

ACCESS TO TRADITIONAL KNOWLEDGE ASSOCIATED WITH GENETIC RESOURCES

In accordance with domestic law, each Party shall take measures, as appropriate, with the aim of ensuring that traditional knowledge associated with genetic resources that is held by indigenous and local communities is accessed with the prior and informed consent or approval and involvement of these indigenous and local communities, and that mutually agreed terms have been established.

NAGOYA PROTOCOL ON ACCESS TO GENETIC RESOURCES AND THE FAIR AND EQUITABLE SHARING OF BENEFITS ARISING FROM THEIR UTILIZATION TO THE CONVENTION ON BIOLOGICAL DIVERSITY

TEXT AND ANNEX



First Nations Information Governance Centre

← → ↺ 🏠

🔒 https://fnigc.ca

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Français 🔍



FNIGC | CGIPN
First Nations Information Governance Centre
Le Centre de gouvernance de l'information des Premières Nations

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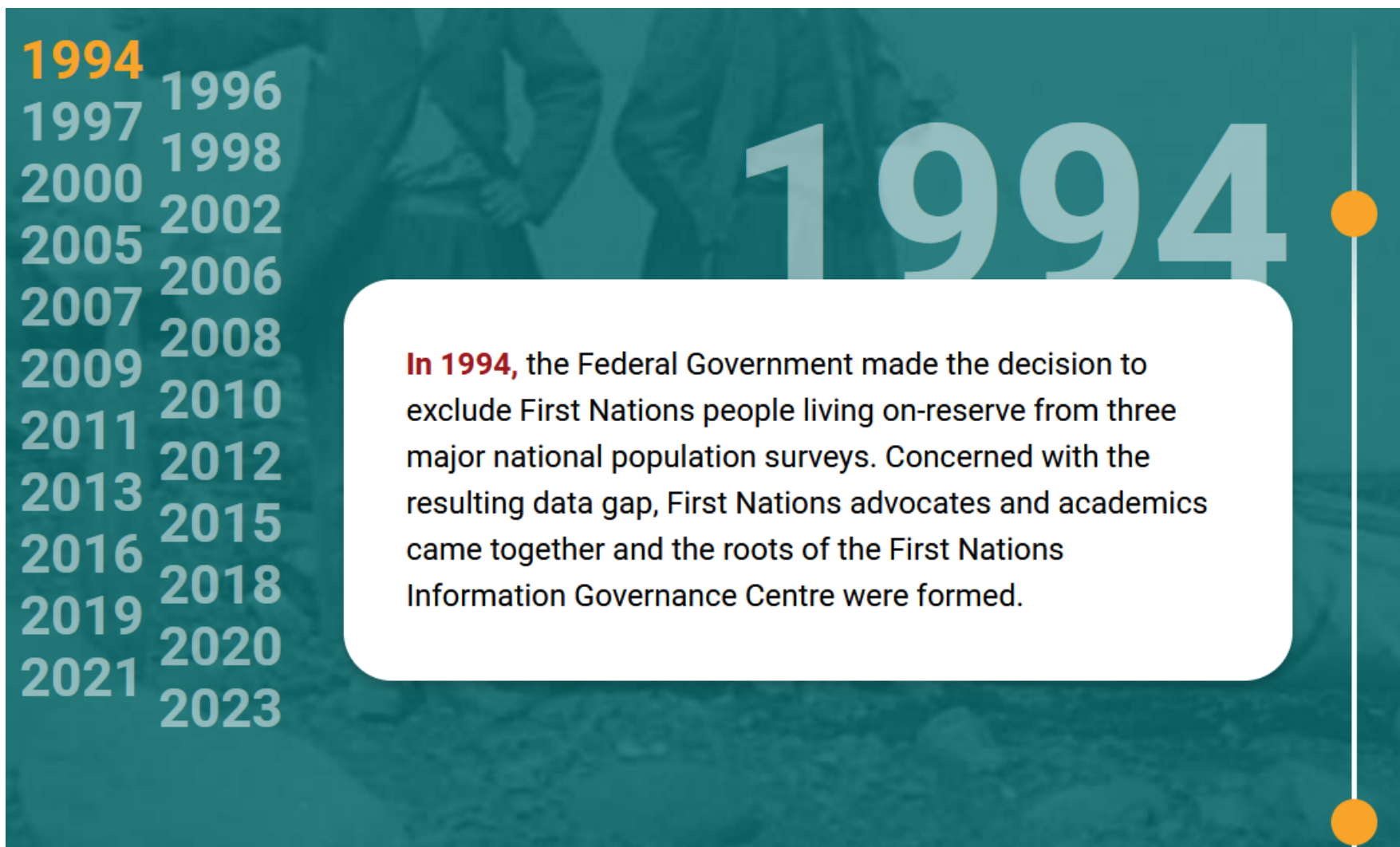
Our data.
Our stories.
Our future.

We envision that every First Nation will achieve data sovereignty in alignment with its distinct world view.

LEARN MORE ABOUT WHO WE ARE



First Nations Information Governance Centre



Governance, Ownership and Ethics



<https://fnigc.ca/ocap-training/>



Welcome to the Indigenous Foods Knowledges Network

Bridging Arctic and US Southwest Indigenous Knowledge systems for food and knowledge sovereignty

Photo: Michael Kotutwa Johnson



About

What is IFKN?



Our Food and Knowledge Stories

View our newest stories



Meetings & Events

Learn more about our past and future meetings and events

In the context of research in general



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1

In case any update is made, the hyperlinks lead to the most recent version of the documents.

In the context of data production

GIDA – Global Indigenous Data Alliance

CARE principles for data (<https://www.gida-global.org>)



TABLE 2 Operationalizing the CARE principles: Tribal expectations for benefit sharing.

CARE principle	Issues raised by communities	Actions for institutions and researchers
Collective benefit	Compensation	Compensation should not be viewed merely as monetary payment for participation in studies, but should be understood to include access to research findings; acknowledgment as author, co-author, or contributor; intellectual property rights; employment and training; and technical assistance.
	Royalties	Tribal entities must be consulted, and contracted with, for the provision of and/or the division of royalties from any products of research.
	Benefits of research generally	The relevance and value of research results to the interests of the Tribe and individual members must be explained and demonstrated.
	Collective benefit	The intent of the research project as well as the benefit(s) that the project, research, or activity will bring to the Tribal community (in addition to researchers and funders) should be clearly explained.
	Lack of benefit	Prior to obtaining permission to conduct studies with Tribes and their members, it should be demonstrated that the project will not be purely extractive, but will be collaborative and inclusive.
	Health	Health-related research should directly benefit the Tribal collective and individual members, and explanations should be given for how study results will be used to improve the health status of the Tribe and its members.
Authority to control	Commercialization	A separate agreement will be made if research outcomes are to be commercialized, as agreed to by the Tribal entity. Restrictions on commercialization or profit can be made.
	Intellectual property	Tools for transparency, integrity, and provenance (e.g., legal agreements, permits, licensing, authorship) should be used.
	Return of/access to findings	Results and findings from research with Tribal entities and communities must first be provided to those nations, communities, and individuals, and the appropriate Tribal entities must be provided with all final reports and data.
	Authorship and acknowledgement	Tribal contributions to the study/project must be acknowledged and properly credited by recognition of individuals and collectives involved in the development, implementation, and return of findings, at the discretion of the Tribe, group, and individuals.
Responsibility	Empowerment	Tribal ideas, values, and voices must be included in the research process, including language and lived experiences.
	Employment/training	Beyond participation in the study, Tribal members should be employed by the project, trained to conduct such research, and be involved in the dissemination of findings.
	Technical assistance	Assistance should be available to the Tribe for writing grants, conducting in-service trainings, developing educational materials, planning Tribal research conferences, and obtaining equipment.
Ethics	Benefit-risk balance	Studies and research projects must present only reasonable risks in relation to anticipated benefits to the Tribe and its members, as determined by the community.

Literature

Mastrandrea, M.D, et al. (2010): Guidance Note for Lead Authors of the IPCC Fifth Assessment Report on Consistent Treatment of Uncertainties. Intergovernmental Panel on Climate Change (IPCC). Available at <<http://www.ipcc.ch>>

Whyte K. (2019): Too late for indigenous climate justice: Ecological and relational tipping points. *WIREs Clim Change*. 2020; 11:e603. <https://doi.org/10.1002/wcc.603> (<https://knowledgecircles.ca/wp-content/uploads/2022/02/Whyte-2019.pdf>)

McDowell et al. (2023): The Canadian Mountain Assessment: Walking Together to Enhance Understanding of Mountains in Canada

Holley C., et al. (2012): The New Environmental Governance, ISBN 9781849714105 , Routledge (2019: https://www.researchgate.net/publication/336922285_A_New_Environmental_Governance)

UN (2007): UNDRIP - United Nations Declaration on the Rights of Indigenous Peoples (https://www.un.org/development/desa/indigenouspeoples/wp-content/uploads/sites/19/2018/11/UNDRIP_E_web.pdf)

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Integrating ILK and conventional science – Sub Saharan Africa

Indigenous and local knowledge especially revolving around forecasting of rain

Regional climate outlook forums (COFs) exist across countries

Need for scientific backup under rapidly changing climate

- Coordination of formal and informal institutions around a central task
- Building communication structures to support forecast dissemination
- Educating the public about forecasts and climate risk
- Better management of seasonal climate variability risk
- Empowering rural communities through participation in forecasting workshops

Mafongoya et al. 2017

Table 1: Indigenous indicators for weather and climate in SSA – tree phenology

Indicator	Country	Significance	Reference
Onset of the rains			
Flowering of the peach tree (<i>Prunus persica</i>), apricot (<i>Prunus armeniaca</i>), budding of acacia species	Botswana Zimbabwe South Africa Tanzania Mali Nigeria	Beginning of rainy season	Kolawole <i>et al.</i> , 2014; Risiro <i>et al.</i> , 2012; Mapfumo <i>et al.</i> , 2015; Zuma-Netshiukhwi <i>et al.</i> , 2013; Kangalawe <i>et al.</i> , 2011; Elia <i>et al.</i> , 2014; Kiptot, 2007;
Flowering coffee, delonix (<i>Christmas</i>) and mango trees; and other trees	Uganda Tanzania Kenya Burkina Faso	Rains are not far	Orlove <i>et al.</i> , 2010; Kangalawe <i>et al.</i> , 2011; Elia <i>et al.</i> 2014; Roudier <i>et al.</i> , 2012; Roncoli <i>et al.</i> , 2002;
Season quality			
Behaviour of certain plants: sprouting of Aloe ferox; germination of new leaves on baobab and tamarind trees	Ethiopia South Africa Zimbabwe Zambia Burkina Faso	Indication of good rains	Saitabau 2014; Speranza <i>et al.</i> , 2009; Zuma-Netshiukhwi <i>et al.</i> , 2013;
Mango tree (<i>Mangifera indica</i>)	Tanzania Zimbabwe Burkina Faso	Heavy flowering of the mango trees indicates a potential drought season	Kijazi <i>et al.</i> , 2013; Elia <i>et al.</i> , 2014; Roncoli <i>et al.</i> , 2002;
Muchakata tree (<i>Parinari curatellifolia</i>), gan'acha tree (<i>Lannea discolor</i>), mushuku tree (<i>Uapaca kirkiana</i>)	Zimbabwe Mali Nigeria	Heavy flowering of the trees indicates a potential drought season	Muguti and Maposa, 2012; Risiro <i>et al.</i> , 2012;
Dormancy breaking in certain trees species e.g. mupfuti (<i>Brachystegia boehmii</i>)	Zimbabwe	Indicates plenty of rain in a few days	Muguti and Maposa, 2012;
Profuse fruiting of certain tree species	Burkina Faso Ethiopia Kenya Nigeria	More fruiting of certain trees indicates a challenging rainy season ahead	Roncoli <i>et al.</i> , 2002; Kiptot, 2007; Davis, 2010;

Sub Saharan Africa

Mafongoya et al. 2017

Table 3: Indigenous indicators for weather and climate in SSA – animal behaviour

Indicator	Country	Significance	Reference
Onset of the rains			
Appearance of red ants, rapidly increasing size of anthills	South Africa Zimbabwe	Good rains are coming	Zuma-Netshiukhwi <i>et al.</i> , 2013; Risiro <i>et al.</i> , 2012;
First appearance of sparrows; flock of swallows preceding dark clouds	South Africa Tanzania Uganda Zimbabwe Nigeria Mali	Rain is at hand and farmers should prepare for above normal rains	Orlove <i>et al.</i> , 2010; Kangalawe <i>et al.</i> , 2011; Elia <i>et al.</i> , 2014; Roudier <i>et al.</i> , 2012; Roncoli <i>et al.</i> , 2002;
Appearance of certain birds e.g. stock Singing, nesting and chirping of certain birds	Uganda Tanzania Zimbabwe Burkina Faso Botswana Nigeria Mali Ethiopia	Rain is at hand and farmers should prepare for above normal rains	Kangalawe <i>et al.</i> , 2011; Orlove <i>et al.</i> , 2010; Elia <i>et al.</i> , 2014; Okonya <i>et al.</i> , 2013; Muguti and Maposa, 2012; Risiro <i>et al.</i> , 2012; Roncoli <i>et al.</i> , 2003; Roudier, 2012; Kolawole <i>et al.</i> , 2014;
Termite appearance (<i>Ancistrotermes spp</i>)	Tanzania Uganda Zimbabwe	Appearance of many termites indicate near rainfall onset	Kijazi <i>et al.</i> , 2013; Elia <i>et al.</i> , 2014; Okonya <i>et al.</i> , 2013; Muguti and Maposa, 2012;
Bee eater (<i>Merops hirundineus</i>)	Tanzania Nigeria Mali Botswana	Appearance in October indicate imminent rainfall and/or onset	Kijazi <i>et al.</i> , 2013; Elia <i>et al.</i> , 2014;
Frogs in swampy areas croaking at night	Uganda Zimbabwe	Indicator for onset of rains	Okonya <i>et al.</i> , 2013; Muguti and Maposa, 2012;
Rock rabbit	Zimbabwe	Its unusual squeaking indicates imminent rainfall	Muguti and Maposa, 2012; Risiro <i>et al.</i> , 2012;
Cicadas (nyenze), day flying chafers (mandere), dragon flies (mikonikoni)	Zimbabwe	Appearance of these signifies imminent rainfall	Muguti and Maposa, 2012; Risiro <i>et al.</i> , 2012;
Season quality			
Grunting pigs indicate low humidity	South Africa	Rains are near	Zuma-Netshiukhwi <i>et al.</i> , 2013;
Calves jumping happily	Uganda South Africa	Good rain season	Okonya <i>et al.</i> , 2013; Zuma-Netshiukhwi <i>et al.</i> , 2013;

Sub Saharan Africa

Table 5: Complementarity between science and traditional seasonal climate forecasts (adapted from Moller et al., 2004)

Principle	Explanation
Diachronic vs systematic complementarity	Science collects dynamic data in a time series over a large area e.g. the use of COFs in Africa. IKS focus on diachronic data and long term series' in small areas e.g. the use of indigenous indicators of forecasting in African villages or districts.
Focus on averages vs extremes	Science is based on the numerical data analysis of averages. IKS observe extreme weather events and unusual patterns, which is critical in climate change adaptation.
Quantitative vs qualitative information	Science demands quantitative data. IKS demand qualitative understanding of the system. Understanding complex systems like climate change requires both. Qualitative measures are rapid and inexpensive but they lack precision.
Hypothesis vs mechanisms	IKS rely on hypotheses for problem solving. Science has powerful tools to explain the reasons for mechanisms. The use of both IKS and science takes advantage of their relative strengths.
Objectivity vs subjectivity	Science is objective and excludes people's feelings. IKS include people's feelings, relationships and sacredness. The combination of both methods produces science with a heart.

Climate knowledge reflected in language

Take aways for scientific knowledge from available local knowledge

Granularity of information not provided in scientific knowledge (e.g. value, quality, timing of rainfall)

No link between knowledge and responsibility in scientific knowledge

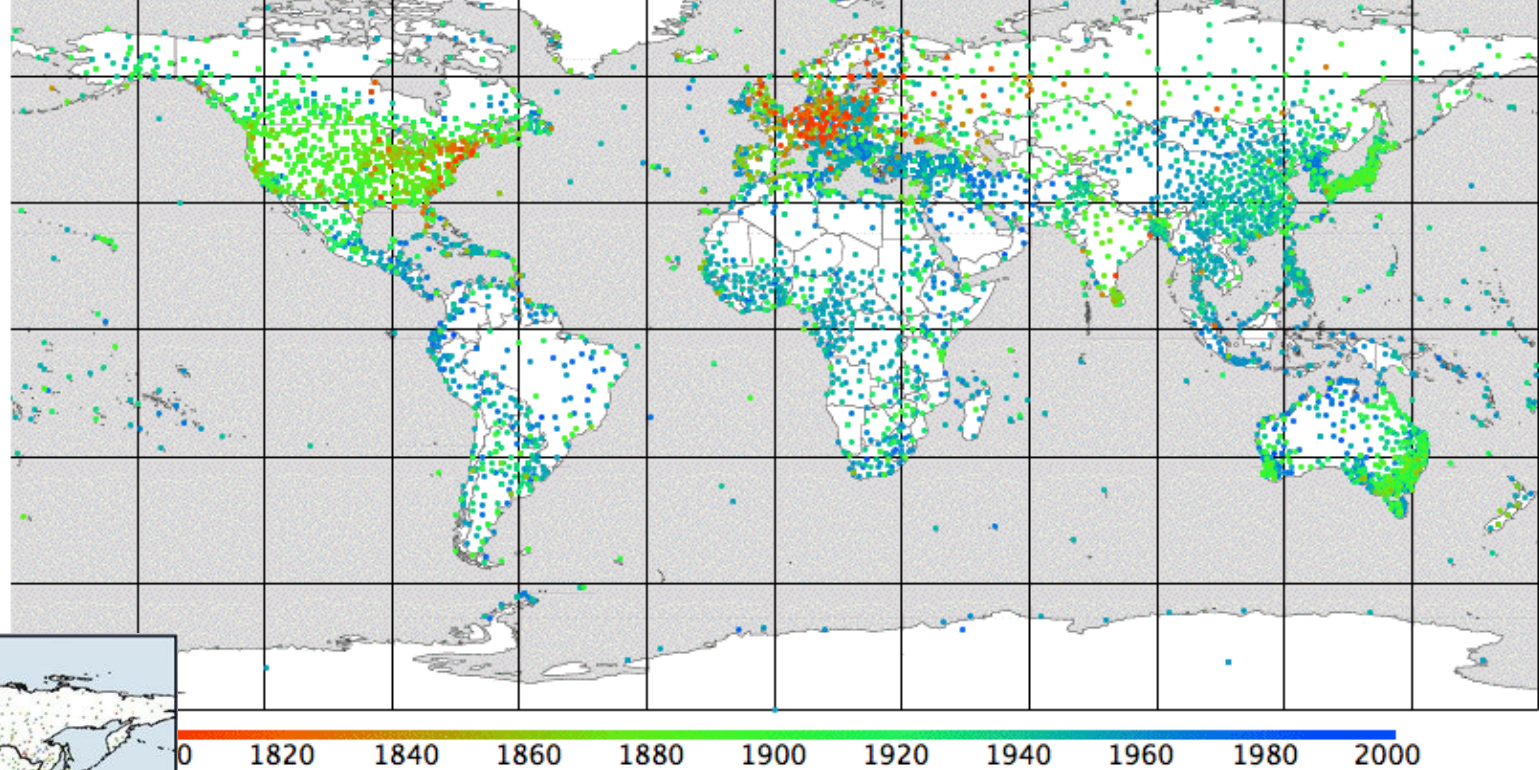
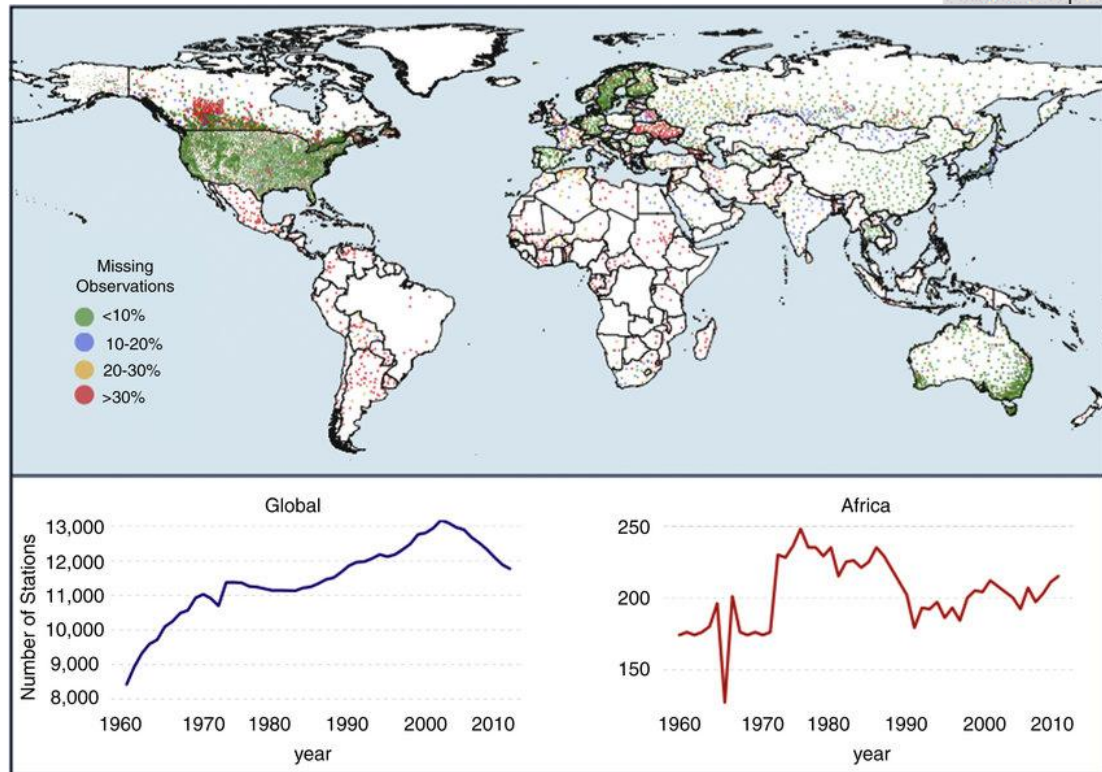
Roncoli et al. 2002

TABLE 2 Typology of Rainfall Events

<i>sa kenga</i>	Big rain. Refers to amount, duration, or distribution.
<i>sa serdem</i>	Fine rain, lasting several hours. Causes good infiltration
<i>sigri saaga (sig saaga)</i>	Planting rain, rain that marks onset of rainy season. Lasts the whole night: 6–8 hours. Steady precipitation, not violent, without wind or thunder. Leaves soil moist for several (5+) days.
<i>sa sika</i>	Very localized, with one or few clouds. Without wind or cool temperatures. Can fall any time of day, but mostly in August.
<i>sa nyanga</i>	Abundant precipitation, mostly mornings or evenings. Lasts several hours. Accompanied by gentle thundering. Leaves soil soaked, good infiltration, moisture lasts long time. Characterizes favorable rainy season and predicts good harvest. Mostly in July–August, but increasingly rare in recent years.
<i>saraogo</i>	Short but violent downpour, usually early morning or at sunset. Localized or oddly distributed. Accompanied by heavy wind, sharp thunder, and lightning. Damages animals, people, dwellings, crops. Believed to be invoked or attracted. Characterizes unfavorable rainy seasons and predicts poor harvest. Mostly in July–August, increasing frequency in recent years.
<i>bind saaga</i>	Literally “excrement rain” because it washes dried excrement off plants in the bush. Also called “mango rains” because it washes mango fruits. Early rains in April. Abundant in the past, used to fill stream beds. Enables people to use mud to repair walls. Stimulates growth of young fibers of roofing grass or <i>baganá (Philostigma sp.)</i> .

Rainfall data

WMO / GHCNd network



Heft Neal et al. 2017

Filling the gap with citizen science



About Us

News

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Projects and Partners

Weather stations and Climate data

Publications

Lela Primary School (TA00001) Kenya

Surface air temper...

Last month

Surface air temperature Lela Primary School (TA00001 - Kenya) Last month



35 °C
30 °C
25 °C
20 °C

20-Jan 22-Jan 24-Jan 26-Jan 28-Jan 30-Jan 1-Feb 3-Feb 5-Feb 7-Feb 9-Feb 11-Feb 13-Feb 15-Feb 17-Feb 19-Feb

Click and drag in the chart to zoom in and inspect the data.

Acquire African Weather Data

Type and press enter to search

EPIC Africa Stakeholder Engageme...



About TAHMO

The Trans-African Hydro-Meteorological Observatory (TAHMO) aims to develop a vast network of weather stations across Africa. Current and historic weather data is important for agricultural, climate monitoring, and many hydro-meteorological applications.

Watch the video below to learn more about our project



NON-TRADITIONAL WATER MONITORING THROUGH LOCAL INVOLVEMENT

Generate, manage and exchange key water data and knowledge through
local involvement for sustainable and effective water outcomes
in the developing context

Citizen Science Nepal

<https://smartphones4water.org/about/>

Mustang, Nepal
October 2023



Astronomical forecasting

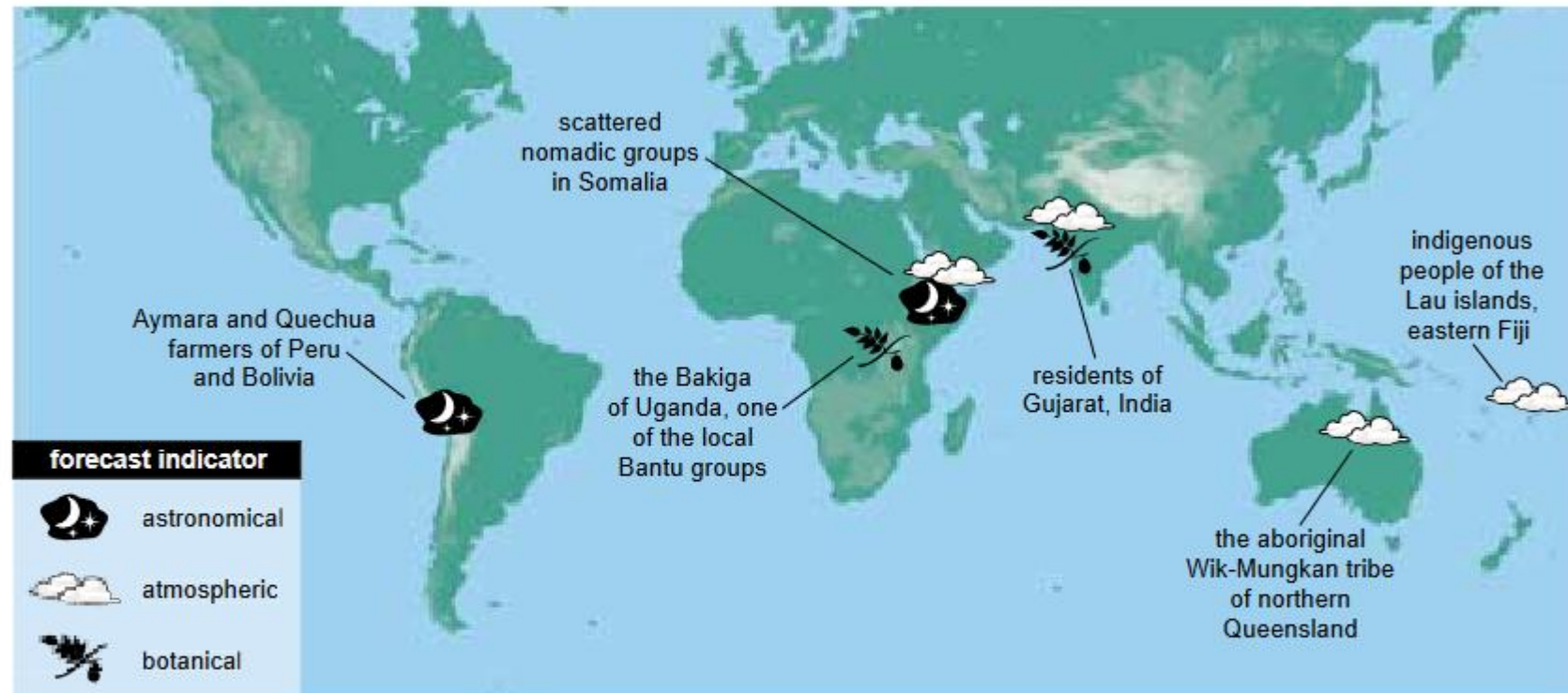


Figure 9. Local and indigenous groups in several parts of the world attempt to forecast climate conditions using simple observations of their environment. Those instances depicted here have recently come to the attention of the authors, who are assembling information about such practices worldwide. Curiously, such predictive schemes appear to be most common in tropical latitudes. Where other examples exist and how much such folk wisdom can be counted on remain open questions. In any event, these forecasts demonstrate which aspects of the climate their practitioners hope to discern and when reliable predictions would be of value to them.

Astronomical forecasting (Andes)

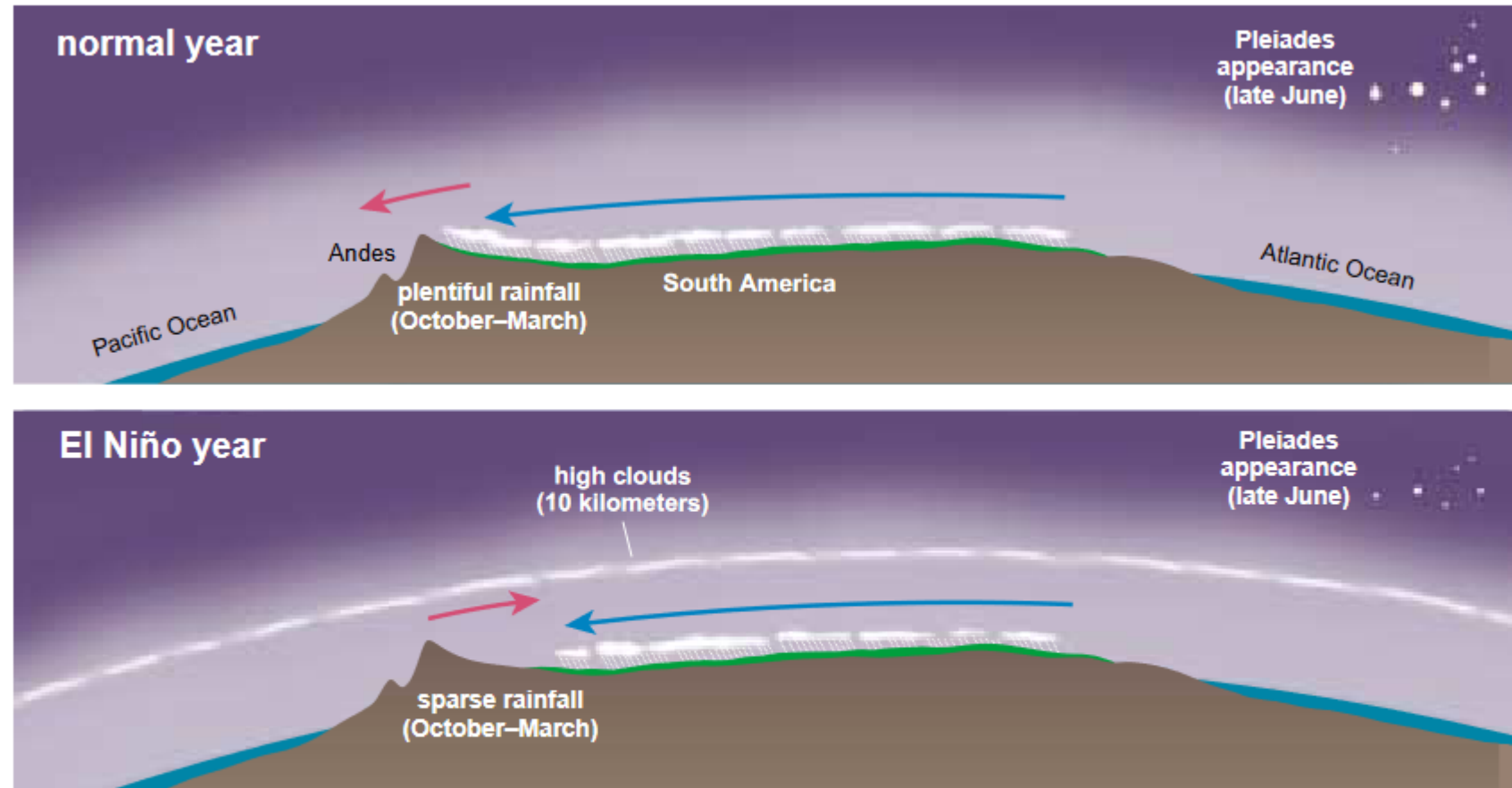


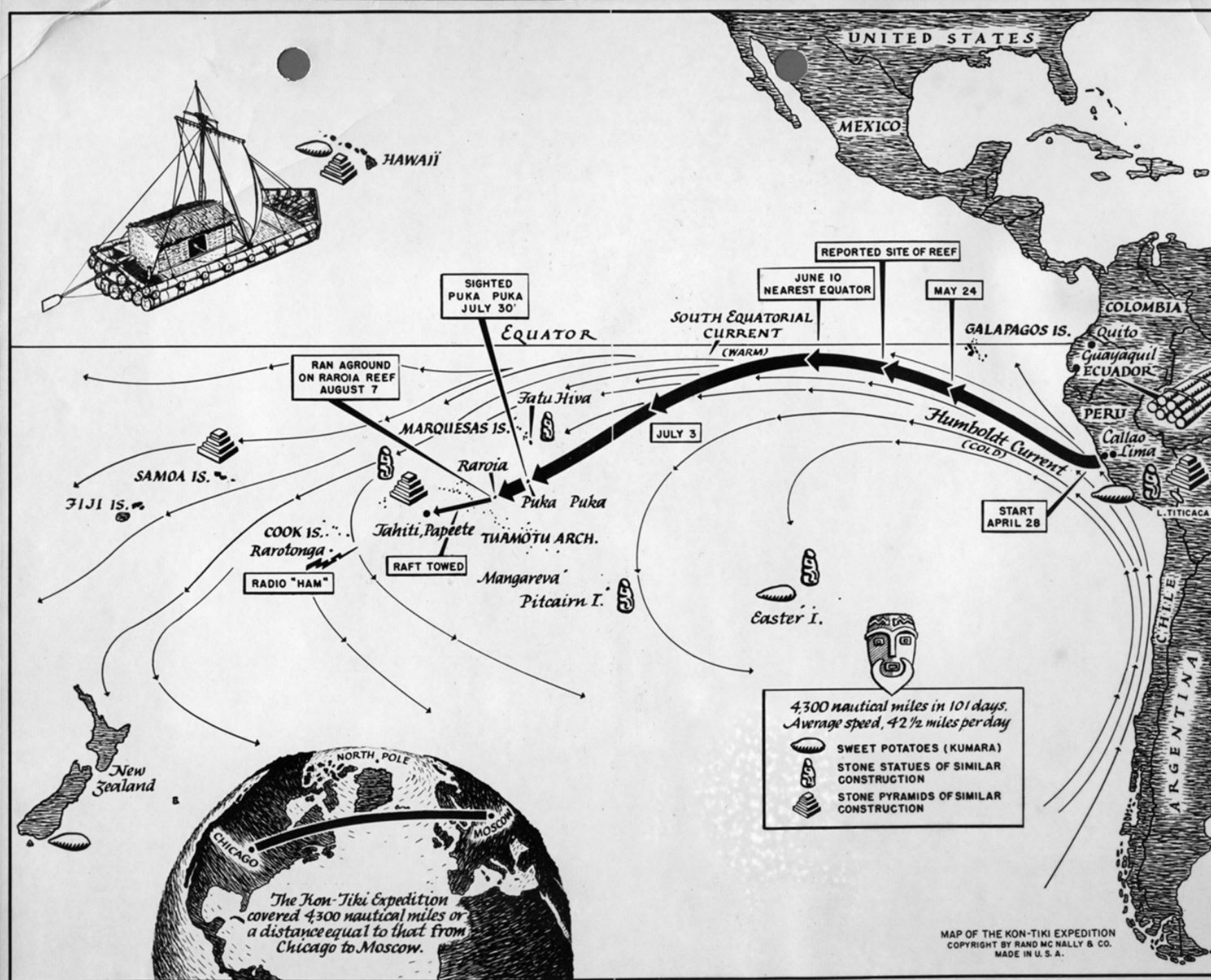
Figure 6. Although the high-level summer winds passing over the Andes are variable, in a normal year (*top panel*) this flow is, on average, from east to west (*red arrow*). These winds thus help to bring some of the moist air moving west from the northern Amazon basin and then south along the flanks of the mountain chain (*blue arrow*) up into the Andes proper. El Niño causes the high-level summer winds to blow, on average, from west to east, which inhibits the import of this humid air and creates a markedly drier season for the farmers of the region (*bottom panel*). They are, however, able to forecast the amount of rain to expect during the summer using the midwinter appearance of the Pleiades as a guide: Because El Niño also creates an obscuring layer of thin, high cloud over much of the tropics, the Pleiades appear dimmer when dry conditions are in store.

Thor Heyerdhal's Kon Tiki

Failed attempt to disprove population of Pacific islands from Asia

Assumption that local knowledge could not explain ocean currents

Assumption that navigation needs to be based on maps



Have a look at the [Canadian Climate Atlas](#) (and for more resources the [main page](#)) as well as a section of the [Canadian Mountain Assessment](#), especially the Introduction and the [Table of Contents](#) and choose 1 or 2 sections that catch your eyes. After reading and clicking your way through, reflect on the questions shown below, and then submit your reflection as an online text here. You will have an opportunity to discuss your answers with students in class too.

- 1) The way knowledges are combined here is different, with different aims and regions, but the underlying motivations are similar. Do you think this way of getting knowledges and the associated processes across is helpful for an outsider to comprehend what matters and what is going on? What do you think can be improved?
- 2) Compare this to the 'drier' assessments you have seen before (like IPCC, IPBES, etc.). Do you see advantages/disadvantages of different approaches? What do you think sticks with policy makers? What sticks with the public? And why?
- 3) Could you imagine this being done for Germany? Would it be something the public or policy makers here would appreciate?

Literature

Mafongoya, P.L. et al. (2017): Chapter 3: Using indigenous knowledge for seasonal quality prediction in managing climate risk in sub-Saharan Africa. https://publications.cta.int/media/publications/downloads/2009_PDF.pdf

Orlove, B.S., Chiang, S., John, C.H. and Cane, M.A. 2002. 'Ethnoclimatology in the Andes.' American Scientist 90: 428–435.

Roncoli, C., Ingram, K. and Kirshen, P. 2002. 'Reading the rains: Local knowledge and rainfall forecasting in Burkina Faso.' Society and Nature Research 15: 409.

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Welcome to our third session 😊

Learning Objectives: *Understand and be able to explain how Indigenous and local knowledge is today included in global assessments of our state of knowledge of the environment and climate and where gaps are. Understand how such assessments are compiled and what larger aim the inclusion of Indigenous and local knowledge could achieve, if it is included.*

Roadmap for today:

- 09:00-10:00 ***Time to prepare your final presentations!***
- 10:00-10:15 **Break**
- 10:15-12:00 Knowledge governance and Ethics
- 12:00-13:00 **Lunch Break**
- 13:00-14:45 ILK and conventional data - Read-Think-Answer (see below)
- 14:45-15:00 **Break**
- 15:00-16:15 Final presentations
- 16:15-16:30 **Break**
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Final presentations

14:45 **Lara Wadlinger, Simon Bertels:** Indigenous Knowledge in River Restoration

14:57 **Benjamin Ouerhani, Joel von Storch:** Shamanism

15:10 **Joseph Mezza, Lara Allgaier, Lisa Rheinlaender:** ILK in palaeogenetics

15:27 **Klara Stier, Paula Schröder:** ILK and climate change in a case study in Uganda

15:40: **Adriane Berisha, Florencia Först, Matilda Martínez Pérez:** Body territory

16:00: **Leonie Karger:** Indigenous Women from the Sixties Storytelling at Winona's Place

16:10: **Laura Uhl, Tim Strobel, Xuan Han Chen:** Indigenous Knowledge for Marine Conservation

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Course Evaluation University



Internal evaluation



SCAN ME

Thank you

